

🍊 Independence vs. Covariance: Relationship Measures

When working with multiple random variables, we need tools to describe how they influence each other. **Independence** is the strongest relationship (or lack thereof), while **Covariance** measures a specific, often weaker, type of relationship (linearity).

1. Independence of Random Variables X and Y

Independence means that the knowledge of the outcome of one variable gives **no information** about the outcome of the other.

A. Formal Definition (Using CDF)

X and Y are independent if and only if their **joint Cumulative Distribution Function (CDF)** is the product of their individual **marginal CDFs** for all possible values (a, b) :

$$F_{X,Y}(a, b) = P(X \leq a, Y \leq b) = F_X(a)F_Y(b)$$

B. Practical Definition (Using PDF/PMF)

This definition is often easier to apply in practice. X and Y are independent if and only if their **joint distribution function** factors perfectly into the product of their **marginal distribution functions**:

- **Continuous (PDF):**

$$f_{X,Y}(x, y) = f_X(x)f_Y(y) \quad \text{for all } x, y$$

- **Discrete (PMF):**

$$P(X = x, Y = y) = P(X = x)P(Y = y) \quad \text{for all } x, y$$

C. Key Result for Expectation

If X and Y are independent, the expected value of their product is the product of their individual expected values:

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

A crucial special case is $E[XY] = E[X]E[Y]$. This property is the key link to covariance (see below).

2. Covariance $\text{Cov}(X, Y)$

Covariance measures the degree to which X and Y **vary together** (co-vary). Specifically, it quantifies the strength and direction of their **linear association**.

A. Definition

Covariance is the expected value of the product of the deviations of X and Y from their respective means:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

B. Intuition and Interpretation

Sign of $\text{Cov}(X, Y)$	Relationship Type	Meaning
Positive	Positive Linear	When X is above its mean, Y tends to be above its mean, and vice versa. They move in the same direction .
Negative	Negative Linear	When X is above its mean, Y tends to be below its mean, and vice versa. They move in opposite directions .
Zero	No Linear	There is no detectable linear relationship between X and Y .

C. Computational Formula

Using the linearity of expectation, the definition simplifies to the most common formula for calculation:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

D. Key Properties

The properties you listed are fundamental:

- **Variance Link:** $\text{Cov}(X, X) = \text{Var}(X)$
- **Symmetry:** $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- **Constants:** $\text{Cov}(cX, Y) = c \cdot \text{Cov}(X, Y)$
- **Additivity (Bilinearity):** $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$

E. The Most Important Connection: Independence $\rightarrow$ Zero Covariance

As derived earlier, if X and Y are independent, then $E[XY] = E[X]E[Y]$. Plugging this into the computational formula for covariance gives:

$$\text{If } X, Y \text{ are Independent} \implies \text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$$

- **Crucial Warning (The Reverse): Zero covariance does NOT imply independence.** If $\text{Cov}(X, Y) = 0$, X and Y are said to be **uncorrelated**. They could still be dependent if their relationship is non-linear (e.g., $Y = X^2$ where $X \sim U(-1, 1)$).

3. Correlation (Standardized Covariance) $\rho_{X,Y}$

Because covariance's magnitude depends on the units of X and Y , it's usually standardized into the **Correlation Coefficient** ($\rho_{X,Y}$):

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- **Range:** Correlation is always between -1 and 1:

$$-1 \leq \rho_{X,Y} \leq 1$$

- **Interpretation:** $\rho_{X,Y} = 1$ means a perfect positive linear relationship; $\rho_{X,Y} = -1$ means a perfect negative linear relationship.

Do you want to proceed with the next major topic, **Convolution (Sum of Random Variables)**?